

P5 — Environmental + Integration + DevOps

SIH 26143 · Detailed 6-Phase Work Plan · 6 owned work packages

ROLE OBJECTIVE

Own environmental forcing, Lagrangian source reconstruction, uncertainty propagation and reproducible deployment.

HOW TO USE THIS PLAN

Each package has four parts: what to build/do, the concrete deliverable, dependencies, and an acceptance condition. The owner is responsible for making the output usable by the next phase—not merely completing a task locally.

PHASE 1 — DATA & PREPROCESSING

Work package 1: Metocean ingestion

Acquire and standardise ERA5 wind and CMEMS current data for selected study windows; align timestamps, spatial grids and units; cache forcing fields; record source/version metadata.

Deliverable: standardised wind/current forcing inputs.

Dependencies / acceptance: Depends on selected study area and time window. Acceptance: forcing can be queried consistently for any test timestamp.

PHASE 2 — SPILL DETECTION

Work package 2: Environmental context layer

Attach wind/current context to detected incidents; record forcing availability and missing-data conditions; flag environmental regimes that may increase interpretation uncertainty.

Deliverable: environmental-context object + quality flags.

Dependencies / acceptance: Depends on metocean ingestion. Acceptance: each reconstruction input has a documented forcing source/time.

PHASE 3 — SOURCE RECONSTRUCTION

Work package 3: OpenDrift/OpenOil modelling

Configure Lagrangian particle simulations using wind/current forcing; implement backward hindcast from detected slick and forward drift forecast; define release/particle settings; capture simulation uncertainty and forcing gaps.

Deliverable: reproducible drift-model workflow producing particle trajectories/corridors.

Dependencies / acceptance: Depends on ERA5/CMEMS data and P1 geometry. Acceptance: same incident/forcing configuration reproduces the stored trajectory result.

PHASE 4 — AIS CORRELATION

Work package 4: Environmental trajectory context

Convert particle output into investigator-friendly source/drift corridors, origin-time candidates and uncertainty bounds; pass these to P4 without converting model output into a claim of exact causation.

Deliverable: reconstruction result schema for AIS correlation.

Dependencies / acceptance: Depends on drift simulation. Acceptance: P4 receives a corridor/time window plus uncertainty, not an unjustified exact source point.

PHASE 5 — EVIDENCE FUSION + UI

Work package 5: Uncertainty + abstention integration

Define what happens when forcing is missing, the source corridor becomes too broad, AIS coverage is inadequate or evidence families disagree; propagate uncertainty and implement explicit insufficient-evidence states.

Deliverable: uncertainty/abstention rules + integrated status signals.

Dependencies / acceptance: Depends on P4 evidence features and P2 fusion model. Acceptance: insufficient evidence produces abstention rather than a forced candidate.

PHASE 6 — VALIDATION & DEPLOYMENT

Work package 6: Deployment + system testing

Containerise services with Docker Compose; configure environment variables and reproducible startup; test service failures, timeouts and resource limits; measure practical prototype latency and document scope.

Deliverable: deployable prototype stack + system test report.

Dependencies / acceptance: Depends on all services. Acceptance: clean environment can start the system and execute the controlled demo workflow.

CROSS-TEAM HANDOFF RULE

No component is considered complete if its output cannot be consumed by the next owner. Preserve IDs, timestamps, coordinate systems, provenance, uncertainty and failure states across every handoff.

Core project rule: AIS gaps, uncertain metocean forcing, ambiguous SAR signatures or weak evidence must increase uncertainty or trigger abstention—not be converted into fabricated confidence.